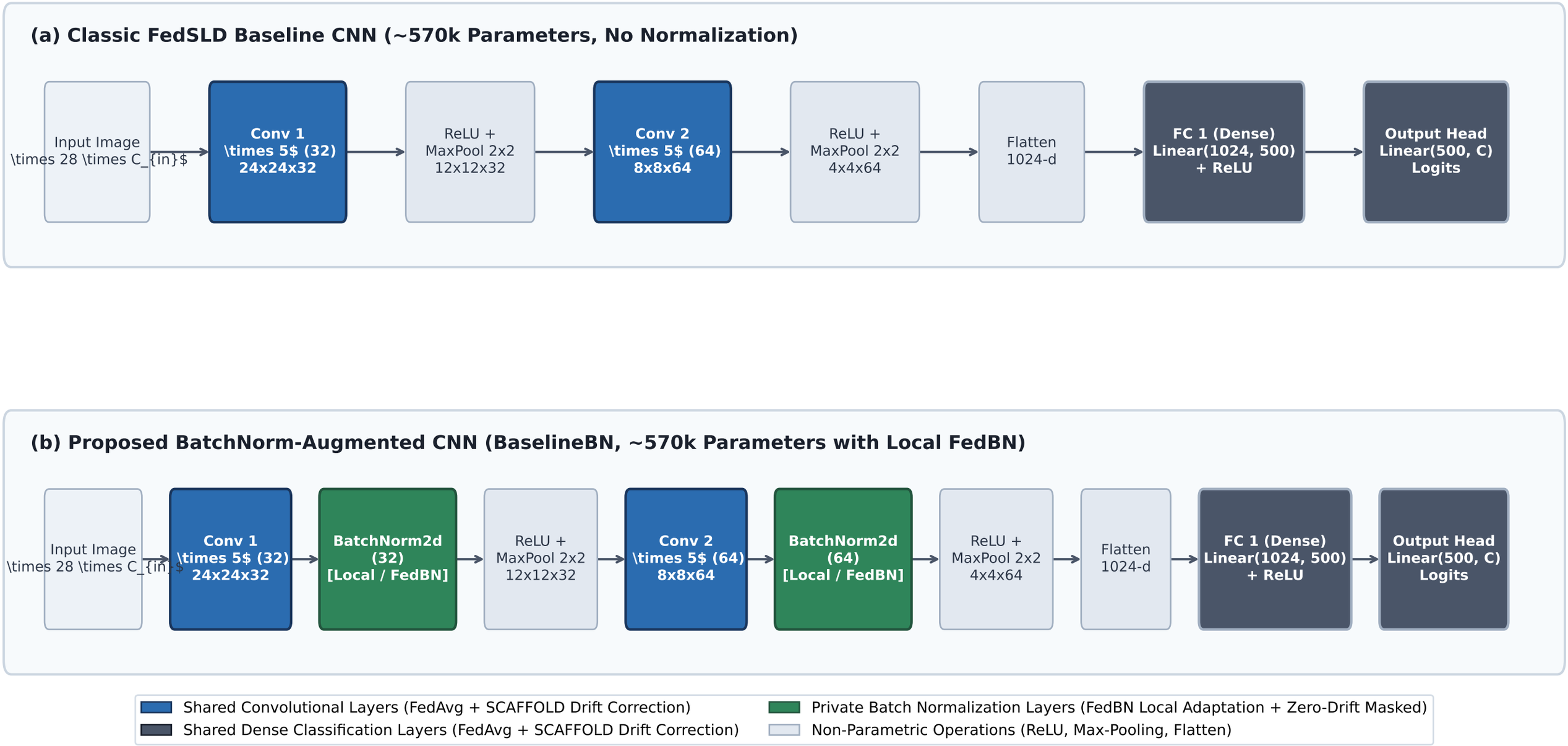
(a) Classic FedSLD Baseline CNN (~570k Parameters, No Normalization)
Input Image \times 28 \times C_{in}$
Conv 1 \times 5$ (32) 24x24x32
ReLU + MaxPool 2x2 12x12x32
Conv 2 \times 5$ (64) 8x8x64
ReLU + MaxPool 2x2 4x4x64
Flatten 1024-d
FC 1 (Dense) Linear(1024, 500) + ReLU
Output Head Linear(500, C) Logits
(b) Proposed BatchNorm-Augmented CNN (BaselineBN, ~570k Parameters with Local FedBN)
Input Image \times 28 \times C_{in}$
Conv 1 \times 5$ (32) 24x24x32
BatchNorm2d (32) [Local / FedBN]
ReLU + MaxPool 2x2 12x12x32
Conv 2 \times 5$ (64) 8x8x64
BatchNorm2d (64) [Local / FedBN]
ReLU + MaxPool 2x2 4x4x64
Flatten 1024-d
FC 1 (Dense) Linear(1024, 500) + ReLU
Output Head Linear(500, C) Logits
Shared Convolutional Layers (FedAvg + SCAFFOLD Drift Correction)
Shared Dense Classification Layers (FedAvg + SCAFFOLD Drift Correction)
Private Batch Normalization Layers (FedBN Local Adaptation + Zero-Drift Masked)
Non-Parametric Operations (ReLU, Max-Pooling, Flatten)